

## PELLO, Finland

WMO: 028440

Lat: 66.773N

Lon: 23.963E

Elev: 86

StdP: 100.30

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -30.8      | -27.9 | -33.6                         | 0.2 | -30.7 | -30.7 | 0.2 | -27.9 | 8.2                   | -4.1 | 7.3 | -4.2 | 1.0                   | 320  | 0.668 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.0                    | 25.5            | 16.8 | 23.5 | 15.7 | 21.5 | 15.0 | 18.4                | 23.1 | 17.1 | 21.5 | 16.0 | 19.8 | 3.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.6                            | 11.9 | 20.0 | 15.4 | 11.1 | 18.5 | 14.4 | 10.3 | 17.5 | 52.3          | 23.2 | 48.3 | 21.3 | 45.2 | 19.8 | 22.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 7.4           | 6.4  | 5.6 | -33.8                      | 27.8 | 2.9                | 2.2 | -35.9                                              | 29.4 | -37.5      | 30.6 | -39.2      | 31.9 | -41.2      | 33.5 |    |    |
| (5)               |      |     | -33.6                      | 19.7 | 2.9                | 1.4 | -35.6                                              | 20.7 | -37.3      | 21.5 | -38.9      | 22.3 | -41.0      | 23.3 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 1.1   | -12.2 | -11.7 | -6.2  | 0.3   | 6.8   | 12.5  | 15.8  | 13.4  | 7.9   | 0.9   | -5.5 | -9.3 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.12 | 7.79  | 7.96  | 5.76  | 4.05  | 4.26  | 3.62  | 3.14  | 3.35  | 3.60  | 4.81  | 6.33 | 7.77 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3663  | 689   | 607   | 503   | 291   | 119   | 16    | 1     | 8     | 83    | 281   | 466  | 599  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 6298  | 947   | 841   | 762   | 541   | 358   | 177   | 90    | 157   | 313   | 539   | 716  | 858  | (9)  |
| (10) |                                                                                  | CDD10.0     | 425   | 0     | 0     | 0     | 0     | 19    | 91    | 181   | 113   | 20    | 0     | 0    | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 18    | 0     | 0     | 0     | 0     | 0     | 3     | 12    | 3     | 0     | 0     | 0    | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 192   | 0     | 0     | 0     | 0     | 8     | 36    | 119   | 29    | 0     | 0     | 0    | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 24    | 0     | 0     | 0     | 0     | 1     | 4     | 17    | 2     | 0     | 0     | 0    | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.5   | 2.3   | 2.4   | 2.7   | 2.7   | 2.9   | 2.8   | 2.6   | 2.4   | 2.5   | 2.4   | 2.3  | 2.3  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 534   | 38    | 31    | 28    | 28    | 35    | 60    | 76    | 63    | 47    | 42    | 47   | 40   | (15) |
| (16) |                                                                                  | PrecMax     | 680   | 67    | 67    | 58    | 61    | 98    | 111   | 128   | 128   | 92    | 94    | 84   | 79   | (16) |
| (17) |                                                                                  | PrecMin     | 399   | 8     | 6     | 6     | 8     | 4     | 7     | 21    | 16    | 12    | 6     | 13   | 8    | (17) |
| (18) |                                                                                  | PrecStd     | 91    | 13    | 15    | 15    | 14    | 19    | 26    | 31    | 28    | 24    | 20    | 20   | 19   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 3.9   | 5.0   | 7.7   | 14.7  | 24.3  | 26.8  | 28.7  | 26.4  | 18.7  | 11.8  | 6.7  | 4.5  | (19) |
| (20) |                                                                                  |             | MCWB  | 2.2   | 2.6   | 3.7   | 7.8   | 13.6  | 16.9  | 18.4  | 18.8  | 13.8  | 9.9   | 5.7  | 2.7  | (20) |
| (21) |                                                                                  | 2%          | DB    | 1.1   | 1.7   | 5.4   | 11.0  | 20.8  | 23.8  | 26.3  | 23.7  | 16.7  | 10.3  | 4.7  | 2.4  | (21) |
| (22) |                                                                                  |             | MCWB  | 0.3   | 0.6   | 2.0   | 5.6   | 12.6  | 14.9  | 17.3  | 16.3  | 12.5  | 8.9   | 3.8  | 1.6  | (22) |
| (23) |                                                                                  | 5%          | DB    | 0.0   | 0.5   | 3.3   | 8.9   | 17.0  | 21.7  | 24.4  | 21.4  | 15.1  | 8.9   | 3.0  | 1.2  | (23) |
| (24) |                                                                                  |             | MCWB  | -0.3  | -0.1  | 0.6   | 4.5   | 10.2  | 13.8  | 16.6  | 15.7  | 12.2  | 7.8   | 2.6  | 0.7  | (24) |
| (25) |                                                                                  | 10%         | DB    | -1.5  | -0.8  | 1.4   | 6.7   | 14.4  | 19.5  | 22.2  | 19.4  | 13.6  | 7.3   | 1.8  | 0.4  | (25) |
| (26) |                                                                                  |             | MCWB  | -1.9  | -1.3  | -0.2  | 2.9   | 8.9   | 12.9  | 15.7  | 14.8  | 11.4  | 6.3   | 1.4  | 0.0  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 2.4   | 3.0   | 4.0   | 8.4   | 15.0  | 18.2  | 20.1  | 19.6  | 15.1  | 10.7  | 6.1  | 3.0  | (27) |
| (28) |                                                                                  |             | MCDWB | 3.6   | 4.9   | 7.6   | 14.3  | 22.1  | 25.0  | 24.8  | 24.3  | 16.9  | 11.1  | 6.5  | 3.8  | (28) |
| (29) |                                                                                  | 2%          | WB    | 0.5   | 0.8   | 2.3   | 6.1   | 13.0  | 16.3  | 18.7  | 17.9  | 13.7  | 9.3   | 3.9  | 1.9  | (29) |
| (30) |                                                                                  |             | MCDWB | 1.0   | 1.5   | 4.9   | 10.1  | 18.8  | 21.3  | 23.5  | 21.5  | 15.3  | 10.0  | 4.4  | 2.4  | (30) |
| (31) |                                                                                  | 5%          | WB    | -0.3  | 0.0   | 1.1   | 4.7   | 11.1  | 14.8  | 17.7  | 16.5  | 12.8  | 7.9   | 2.6  | 0.7  | (31) |
| (32) |                                                                                  |             | MCDWB | 0.1   | 0.4   | 2.5   | 8.5   | 16.3  | 19.7  | 22.5  | 19.8  | 14.5  | 8.9   | 3.0  | 1.1  | (32) |
| (33) |                                                                                  | 10%         | WB    | -1.9  | -1.3  | 0.1   | 3.3   | 9.4   | 13.7  | 16.6  | 15.4  | 11.6  | 6.4   | 1.5  | 0.1  | (33) |
| (34) |                                                                                  |             | MCDWB | -1.5  | -0.8  | 1.3   | 6.2   | 13.6  | 18.2  | 20.9  | 18.4  | 13.3  | 7.3   | 1.7  | 0.4  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.5   | 8.4   | 10.3  | 9.3   | 9.2   | 8.9   | 9.0   | 9.0   | 7.9   | 5.6   | 5.8  | 7.4  | (35) |
| (36) |                                                                                  |             | MCDBR | 6.7   | 6.4   | 10.4  | 12.2  | 14.1  | 12.6  | 12.5  | 12.0  | 9.7   | 6.1   | 4.7  | 5.7  | (36) |
| (37) |                                                                                  |             | MCWBR | 6.4   | 5.7   | 7.7   | 7.5   | 7.2   | 5.4   | 5.1   | 5.8   | 6.0   | 4.9   | 4.4  | 5.3  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 6.8   | 6.3   | 8.6   | 11.4  | 12.5  | 10.3  | 10.2  | 10.0  | 7.7   | 5.2   | 4.4  | 5.6  | (38) |
| (39) |                                                                                  |             | MCWBR | 6.5   | 5.7   | 6.6   | 7.1   | 6.8   | 5.2   | 5.0   | 5.3   | 5.1   | 4.5   | 4.2  | 5.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.197 | 0.263 | 0.296 | 0.329 | 0.342 | 0.346 | 0.357 | 0.341 | 0.314 | 0.286 | 0.344 | N/A  | (40) |      |
| (41) |                                                                                  | taud        | 1.936 | 2.107 | 2.181 | 2.247 | 2.447 | 2.468 | 2.453 | 2.483 | 2.540 | 2.357 | 2.668 | N/A  | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 187   | 568   | 751   | 825   | 858   | 863   | 838   | 816   | 758   | 561   | 208   | 0    | (42) |      |
| (43) |                                                                                  | Edh at Noon | 14    | 52    | 83    | 102   | 93    | 94    | 93    | 81    | 61    | 42    | 15    | 0    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.07  | 0.56  | 2.03  | 3.78  | 4.78  | 5.17  | 4.86  | 3.55  | 1.90  | 0.67  | 0.13  | 0.00 | (44) |      |
| (45) |                                                                                  | RadStd      | 0.00  | 0.09  | 0.19  | 0.25  | 0.47  | 0.40  | 0.46  | 0.29  | 0.23  | 0.12  | 0.02  | 0.00 | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.70 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (45 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page